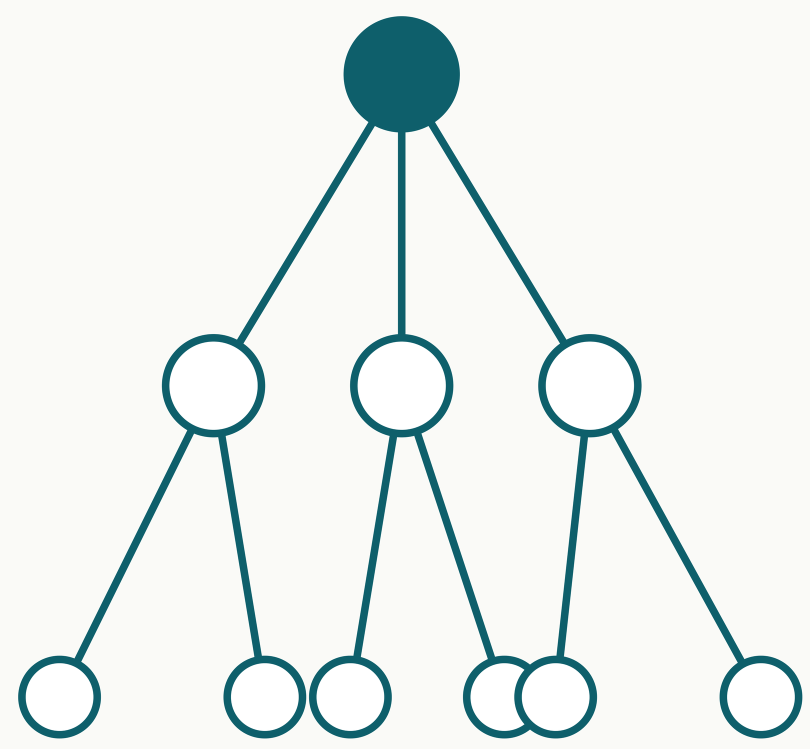
The Bruhat–Tits Tree Code — Status and Test Plan

Rowan Brad Quni-Gudzinas · QNFO · CWI Summer School on Quantum Algorithms & QEC · Wed 26 Aug 2026, 16:30–18:00

Why it matters

Fault tolerance today means surface codes: many hundreds of physical qubits per logical qubit, active syndrome measurement, millikelvin cooling. The tree code is a different geometry whose **simulated** thresholds run 4.6–75× above surface-code figures — and whether that survives a matched-noise comparison is test T1. If it does, the payoff is energy: fewer joules per correct answer.



The Bruhat–Tits tree T_2 — the geometry behind the code. Every vertex carries a [[3,1,1]] perfect tensor.

Verified

Thresholds computed three ways — analytical recursion, exact stabilizer simulation, Monte Carlo — and re-checked by a deposited 35-check script.

10.5281/zenodo.20109836 · 10.5281/zenodo.22038733

Boundary

Under independent errors the qudit family sits at ≈ 2×10^{-4}, about 55× below surface codes. The advantage claim targets correlated-failure regimes — exactly what T1 probes.

10.5281/zenodo.21046993 · 10.5281/zenodo.22025544

Not yet

No hardware run. Tests T3–T4 are the instrumentation path.

The code and the numbers

A holographic perfect-tensor code on a truncated Bruhat–Tits tree: logical information lives at the root, physical qubits at the leaves, and error correction is built into the geometry rather than applied by active measurement loops.

Error model	Tree code	Surface code	Ratio
Bit-flip	**50.0%**	~10.9%	4.6×
Depolarizing	**75.0%**	~1.0%	75×
Independent X+Z	**17.30%**	—	—

Classical simulation only: i.i.d. noise, crossing criterion; surface comparators are the standard published figures (bit-flip ~10.9% code-capacity, depolarizing ~1% circuit-level). No tree code has run on quantum hardware yet.

The tests that would change my mind

T1 · Matched-noise comparison

Tree vs surface under the *same* correlated noise model. Classical, days–weeks.

Would change my mind: the advantage falls to parity.

T2 · Decoder + full overhead

Physical qubits plus classical decoding cost. Classical.

Would change my mind: the advantage disappears once decoding cost is counted.

T3 · Random walk on a p-regular tree

Return-probability decay. Ion / analog simulator, ~8 weeks.

Would change my mind: no change in the decay exponent where predicted.

T4 · Clock-rest split

Diagonal 0% vs non-diagonal 29–35% violation rate. Trapped ion, ~8 weeks.

Would change my mind: the measured split falls outside that range.

T5 · Independent re-run

An outside group repeats T1 with my simulator and the protocol.

Would change my mind: nothing to predict — this one is about reproducibility.

1. A correlated-noise benchmark — model and data — I can run T1 against.
2. A decoder reality check: does a known decoder family already cover tree-structured codes, and how does it scale?
3. Pointers on perfect tensors for p > 2 (the binary [[3,1,1]] case is settled).
4. An independent re-run of T1.

The five tests are the takeaway: every criterion is stated with numbers, so a null result is a result. The simulator and the protocol are handed over to anyone who wants to run T1 or T5 themselves.

All numbers are classical simulation — i.i.d. noise, crossing criterion; nothing has run on hardware yet. · DOIs (verified 21 Aug 2026): 10.5281/zenodo.20109836 (architecture) · 10.5281/zenodo.22038733 (guide + verification script) · 10.5281/zenodo.21046993 (qudit QEC) · 10.5281/zenodo.22025544 (trapped-ion testbed) · 10.5281/zenodo.21964674 (QEC–Darwinism audit) · 10.5281/zenodo.21979060 (prime-valuation correction) · Full plan: github.com/QNFO/cwi-qec-poster-2026 → docs/tree-code-roadmap.md